Branch Coverage Report

Decision-branch coverage of the server code, measured while running the full automated test suite.

1. Summary

Branch coverage (reported)	Branches found	Branches taken	Branches not taken
83.91 %	373	313	60

Scope: server source in src/ (excluding the generated Prisma client). index.ts (process entry point) and seed.ts (data seeding) are not loaded by the tests and are not in the figures. The React client (client/src) has no automated tests and is not covered. Measured with Node's built-in V8 coverage (--experimental-test-coverage).

2. Branch coverage by file

File	Branches	Taken	Not taken	Branch %	Lines with an untaken branch
activityLog.ts	12	11	1	91.67	13
app.ts	9	6	3	66.67	31, 42
attempts.ts	60	53	7	88.33	59, 108, 109, 142, 148, 181, 216
auth.ts	34	29	5	85.29	32, 38, 88, 99, 109
backup.ts	74	63	11	85.14	47, 50, 70, 73, 76, 79, 80, 92, 110, 193, 214
exams.ts	31	27	4	87.10	103, 110, 119, 139
grading.ts	16	16	0	100.00	-
http.ts	31	22	9	70.97	28, 36, 37, 42, 50, 54, 63, 71
prisma.ts	4	2	2	50.00	8, 14
questions.ts	37	36	1	97.30	10
reports.ts	26	18	8	69.23	30, 31, 32, 58, 74, 82
results.ts	22	15	7	68.18	29, 30, 31, 87, 99, 100
routes.ts	1	1	0	100.00	-
security.ts	16	14	2	87.50	33, 41
All files	373	313	60	83.91	

3. Gaps

Weakest branch coverage: prisma.ts (50%: the missing-DATABASE_URL error and optional-port fallback), app.ts (66.7%), results.ts (68.2%) and reports.ts (69.2%: empty-data and optional-filter paths), and http.ts (71.0%: error-mapping cases). grading.ts, routes.ts and questions.ts are at or near 100%. The branch figure of 83.9% is lower than line coverage (94.4%) because most untested code is defensive fallback/else paths. Adding tests for empty reports/results, optional query filters and malformed requests would close most of the gap.

4. Reproduce

Same command as the Code Coverage Report; add --test-reporter=lcov --test-reporter-destination=lcov.info to obtain per-branch (BRDA) data.